

Predicting entrepreneurial pitch success using behavioral, textual, and EEG neuroforecasting

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Less than 2% of US capital is attracted by women-owned businesses. Closing this gap would be facilitated by a deeper understanding of how entrepreneurial pitches engage investor interest and convert them to funding. Traditional methods primarily rely on self-reported subjective assessments, which may not fully capture how a pitch shapes investor decisions in real-time. Here, we bridge this gap by applying a neuroforecasting approach that integrates self-reports, textual sentiment analysis, and neural responses via EEG to forecast the outcomes of entrepreneurs' pitches. We conducted an EEG experiment with 28 subjects who watched 14 Shark Tank pitches, then analyzed EEG frontal alpha asymmetry (FAA)—a biometric measure of positive motivation—from power spectral density and neural synchrony (intersubject correlation)—a measure of shared attention, emotion, and mindset—from time-frequency representations. We also employed natural language processing to analyze transcripts for various sentiments. After each pitch video, participants self-reported their interest in obtaining further information, willingness to invest, perception of the investment's quality, and expected financial returns. We also measured individuals' tendencies toward dynamic capability (seeking and sensing investment opportunities) via a post-study survey. We first used hierarchical linear models to predict investor interest, with predictors including self-reported pitching content quality, investors' dynamic capabilities, and FAA (averaged whole span of each pitch). We found that all measures significantly predicted investor interest. Importantly, incorporating FAA and dynamic capability measures into the model significantly improved predictions, even controlling for age and gender. However, adding the joyful sentiment from text analysis did not increase variance. Second, the FAA predicted out-of-sample investors' interest (N=497) within the first 10 seconds of pitching, thus providing a biomarker of early impression and interest. Moreover, peak neural synchrony around 50 seconds into the pitch, when entrepreneurs were typically describing their product solutions, significantly predicted real-world deal outcomes. Thus, integrating EEG data with self-reported measures provides a robust predictive approach to evaluating and refining pitches to maximize successful outcomes. Our findings build upon and extend recent work in neuroforecasting predicting pitch success using portable EEG with a small sample, thus offering a practical and scalable solution for entrepreneurs to refine their presentation strategies and successfully secure a deal.

Keywords: neuroforecasting, entrepreneurship, FAA, synchrony, EEG

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